

Numerical Analysis chapter 1 programming report

Zhouyue Qian 3220104074 [†]

(Information and Computational Science Class 2201), Zhejiang University

Due time: September 21, 2024

Abstract

This is the programming homework of chapter 1, which includes a equation solver. I designed a base class for a solver, from which the bisection, Newton, and secant methods are derived. This is specifically accomplished through overloading and inheritance.

A-Description

The program is designed to solve the equation $f(x) = 0$ using the bisection, Newton, and secant methods. The base class abstracts the solution function and the destructor of the function.

```
class EquationSolver {  
public:  
    virtual double solve() = 0 ;  
    virtual ~EquationSolver() = default;  
};
```

As for the three specific methods, they are directly inherited from the base class, and the functions use C++ pointers.^[1]

```
double (*f)(double); //函数  
double (*df)(double); //一阶导数
```

B

After the program is run, the following results are obtained:

^{*}Translator for some paragraph^[3]

[†]Electronic address: 3220104074@edu.zju.cn

```

solution is 0.860334
f(x) is -4.94698e-08
solution is 0.641186
f(x) is -5.61904e-08
solution is 1.82938
f(x) is 3.99701e-08
solution is 0.117877
f(x) is -4.64412e+07

```

Figure 1: results of B

We could find that the errors of the first three zero points are very small. Nevertheless, the error of the fourth zero point is relatively large. The function values are very large. It is obvious that this zero point is not a real zero point. So to find out why this happens, I have drawn the function graph.^[2]

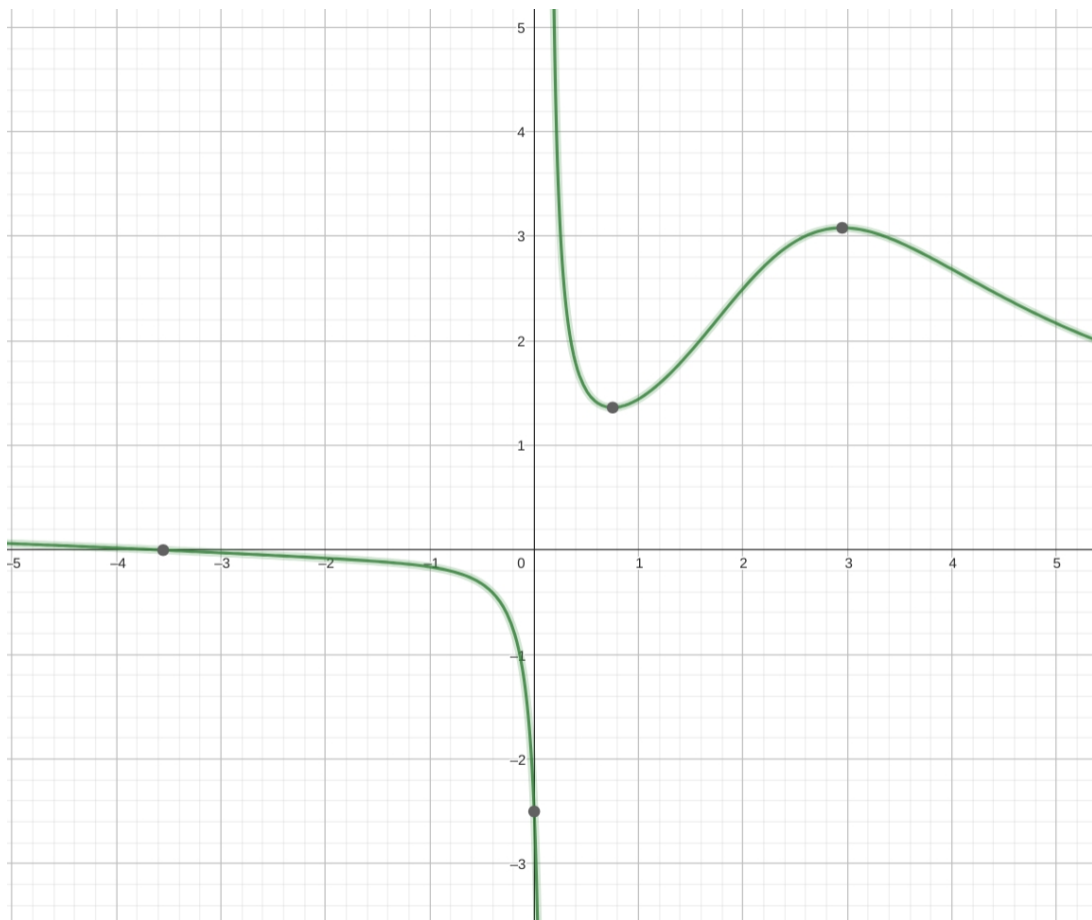


Figure 2: graph of function4

From the image, we can observe that the function is not continuous near $x = 0$. We designate this point of discontinuity as α . Therefore, the data we ultimately obtain through the bisection method is the result of breaking out after the interval length becomes smaller than δ , and this result approaches α . Since the function does not meet the criteria for the bisection method, we cannot obtain the exact zero point.

C

```
solution is 4.49341
f(x) is -5.66529e-08
solution is 7.72525
f(x) is 4.94653e-08
```

Figure 3: results of C

We could find that the errors of the zero points are very small, which means we have already found the exact zero points.

D

```
solution is 3.14093
f(x) is -5.45197e-08
solution is 1.30633
f(x) is -6.52234e-12
solution is -0.188685
f(x) is -6.31696e-09
```

Figure 4: results of D

We could find that the errors of the zero points are very small, which means we have already found the exact zero points. However, plotting the graph of the function, I find that these functions has many zero points.

x_0	x_1
-3π	-3.5π
4	5
11	12

```

solution is 0.166166
f(x) is -5.25141e-08
solution is 0.166166
f(x) is 4.04985e-08
solution is 0.166166
f(x) is 8.16947e-12

```

Figure 6: results of E



Figure 5: graph of function1

So, I change x_0 and x_1 to find different zero points.

E

We could find that the errors of V are very small, which means we have already found the exact zero points. This three zero point is generated by the three methods.

F

a,b

The results have shown in the pics.

```

solution is 32.9722
f(x) is 5.75226e-09
solution is 33.1689
f(x) is 5.2601e-09
solution is -11.5
f(x) is -5.85839e-08
solution is 216867
f(x) is 9.62075e-12

```

Figure 7: results of F

c

We find that if we have totally different zero point in a . Below is the graph of the function.

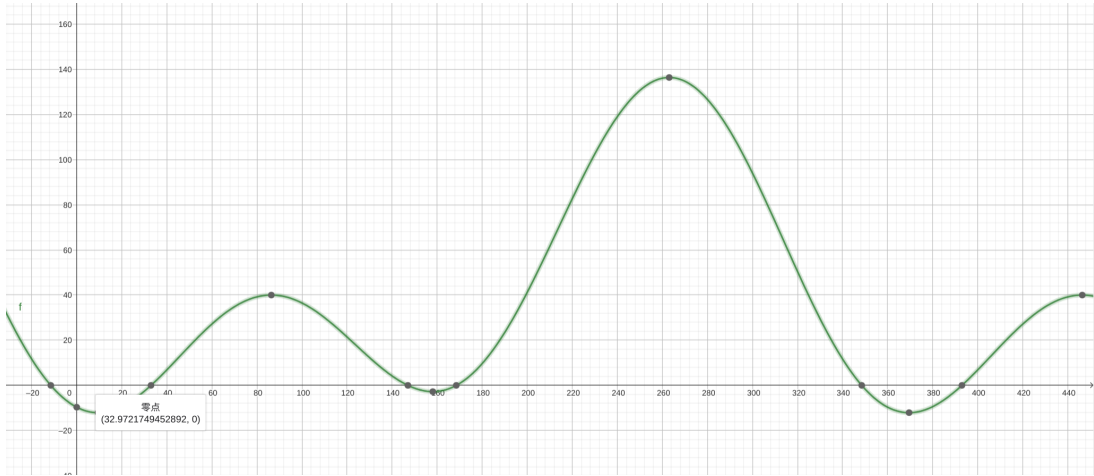


Figure 8: graph of function

[2] From the graph, we could tell that $f(x)$ has tons of zero points. The function is monotonically increasing in the domain where x equals 33. Moreover, the function's concavity remains essentially unchanged.

- if the initial value is taken near 0, since this part is monotonically decreasing and the concavity remains unchanged, it will iterate to a zero point on the left side of 0.
- However, when taken near 90, due to the distance from the zero point being relatively far and the absolute value of the secant slope being very small, after continuous iteration, the zero point will eventually converge to another zero point that is farther away from the initial value.

Acknowledgement

References

- [1] DeepSeek AI. DeepSeek. <https://www.deepseek.com/>. Accessed: 2024-09-19.
- [2] GeoGebra. GeoGebra. <https://www.geogebra.org/graphing>. Accessed: 2024-09-19.
- [3] Moonshot AI. Kimi Assistant. <https://kimi.moonshot.cn/>. Accessed: 2024-09-19.